

Final Report: Newton–Schulz Schedule Design in Muon for Transformer Pretraining

Yicheng Zhao, Sihua Zhang, Ruilin Gao
COE202 Mathematical Foundations of AI, Spring 2026

1 Problem Setup

Problem statement Muon is a recently proposed optimizer that updates matrix parameters by orthogonalizing a momentum-like update direction through a short Newton–Schulz (NS) iteration [2, 3]. Recent theory has also started to analyze the convergence properties of Muon-style NS updates [4]. In practice, the iteration is controlled by quintic coefficients, so schedule design becomes a spectral design problem rather than a minor implementation detail. This project asks: *under a fixed transformer pretraining stack, how do different NS coefficient schedules change the spectral geometry of Muon updates, and how does this spectral geometric change affect validation loss and training cost?*

Mathematical formulation To understand Muon, it is helpful to start from the limitation of AdamW. For a weight matrix $W \in \mathbb{R}^{m \times n}$, AdamW treats the gradient elementwise. This is effective, but it ignores the fact that W is a matrix with coupled row and column geometry. In particular, an update can have very uneven singular values, meaning that some directions are stretched much more strongly than others. Muon is motivated by the idea that for hidden-layer matrix parameters, a better update should acts more isotropically on the singular directions and avoids concentrating into a few dominant directions.

For a rectangular matrix X , X is semi-orthogonal if:

$$X^\top X \approx I_n \quad \text{if } m \geq n, \quad XX^\top \approx I_m \quad \text{if } m < n. \quad (1)$$

Equivalently, if $X = U\Sigma V^\top$ is an SVD, then the ideal semi-orthogonal factor is the polar factor $UV^\top$, whose nonzero singular values are all equal to 1. Muon utilizes an iterative algorithm called NS iteration instead of SVD to approximately orthogonalize the update matrix before applying it to the parameters.

Why can NS iterations produce a semi-orthogonal matrix? The reason is that they act on singular values. After a normalization step, Muon repeatedly applies a quintic matrix polynomial. If X_k is the matrix after the k -th NS iteration, then

$$X_{k+1} = a_k X_k + b_k (X_k X_k^\top) X_k + c_k (X_k X_k^\top)^2 X_k, \quad (2)$$

or the aspect-ratio-swapped equivalent when the matrix is tall. Because this map is equivariant to the SVD, it leaves singular vectors unchanged and transforms only the singular values. At the scalar level,

$$\sigma_{k+1} = p_k(\sigma_k), \quad p_k(\sigma) = a_k \sigma + b_k \sigma^3 + c_k \sigma^5. \quad (3)$$

Therefore, a coefficient schedule $\{(a_k, b_k, c_k)\}_{k=1}^T$ defines a composed spectral map on singular values. If the composed map pushes the relevant singular-value range toward 1, then the resulting matrix becomes closer to the polar factor and hence closer to semi-orthogonality. This is the mathematical core of Muon: orthogonalization is implemented indirectly by shaping the singular-value dynamics with a short polynomial iteration. The high level of a muon update procedure can be summarized as in Algorithm 1.

Algorithm 1 Muon update procedure

- 1: Initialize momentum state M_t
 - 2: **for** $t = 1, 2, \dots$ **do**
 - 3: Compute matrix gradient $G_t = \nabla_W \mathcal{L}_t(W_{t-1})$
 - 4: Update accumulated momentum $M_t \leftarrow \mu M_{t-1} + G_t$
 - 5: Apply NS iteration on M_t to obtain O_t
 - 6: Update parameters: $W_t \leftarrow W_{t-1} - \eta_t O_t$
 - 7: **end for**
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This perspective also explains why coefficient choice matters. Different coefficients create different scalar maps p_k : some are conservative and stable, some are more aggressive and rapidly amplify small singular directions, and some are designed adaptively from a presumed lower bound on the spectrum [1]. In other words, Muon is not just one optimizer; it is a family of optimizers parameterized by spectral maps. In this project, we systematically compare four coefficient schedules of Muon along with a AdamW baseline:

- **AdamW**: no NS orthogonalization on matrix parameters.
- **Stable**: conservative mapping coefficient $(2.0, -1.5, 0.5)$.
- **Manual**: fast-to-stable schedules
- **Fast**: aggressive mapping coefficient $(3.4445, -4.7750, 2.0315)$.
- **Polar Express (PE)**: dynamically compute coefficient at each step [1].

2 Technical Approach

Experimental design The philosophy of the experiment is to isolate the effect of the NS coefficient schedule itself. For that reason, we use a controlled design in which the architecture, data source, token budget, learning-rate schedule, batch structure, and all Muon hyperparameters other than the orthogonalization schedule are held fixed across runs. Under this design, observed differences can be interpreted as consequences of spectral-map design rather than of unrelated tuning changes.

AdamW is included as an external baseline because it provides a useful reference point for what happens when matrix parameters are not orthogonalized. Within the Muon family, we fix the NS iteration depth to $T = 5$ steps for the core schedule comparison. We further conduct several ablation studies on important Muon hyperparameters, especially the iteration depth and the PE lower-bound parameter. Finally, we combine the training results and mathematical analysis of the NS iteration to better interpret the relation between semi-orthogonal updates and validation loss in language pretraining.

Evaluation dimensions The study evaluates each method along three dimensions. First, we measure *optimization quality* through validation-loss trajectories, final validation loss, and token-normalized area under the validation-loss curve (AUC). Second, we measure *computational cost* through end-to-end wall-clock time, because a schedule that improves geometry but is too expensive would be less practically useful. Third, we measure *spectral geometry* directly by analyzing update matrices during training. This geometric analysis is essential: without it, any difference in final loss would remain a black-box empirical observation rather than a mathematically interpretable effect of the NS map.

Experimental setup All experiments are run on one RTX 4090 GPU in BF16 precision using FineWeb-10B tokenized shards [5]. The model is a standard 11-layer pre-norm Transformer with model dimension 768, 6 attention heads, head dimension 128, MLP ratio 4, and sequence length 2048. The total training budget is fixed at 100M tokens, with 131,072 tokens per optimizer step, 16 gradient-accumulation steps, 2% warmup, and cosine decay to 20% of the peak learning rate.

The optimizer settings are chosen to match a realistic small-transformer pretraining regime:

- AdamW baseline: learning rate 6×10^{-4} , $(\beta_1, \beta_2) = (0.9, 0.95)$, weight decay 0.1.
- Muon family: base learning rate 0.023, momentum 0.95, $\beta_2 = 0.9$, weight decay 1.2.

These values lie in the empirically established range commonly used in modern small-transformer pretraining practice and in public Muon baselines [3], so they should be viewed as reasonable default settings rather than arbitrary picks.

Muon variant used in this project The project does not use the simplest momentum SGD shown in Algorithm 1. The actual matrix update path is a *Nesterov-style* lookahead variant together with second-moment rescaling on the , so more precisely it is a normalized Muon implementation. Let G_t denote the matrix gradient at step t , and let B_t be the momentum buffer. The update path is

$$B_t = mB_{t-1} + (1 - m)G_t, \quad (4)$$

$$g_{\text{pre},t} = mB_t + (1 - m)G_t, \quad (5)$$

$$g_{\text{post},t} = \mathcal{P}(g_{\text{pre},t}), \quad (6)$$

where $\mathcal{P}$ is the schedule-defined NS orthogonalization map. The use of $g_{\text{pre},t}$ means the optimizer does not orthogonalize the raw buffer itself, but a Nesterov-style lookahead combination that effectively looks one step ahead along the momentum direction.

After orthogonalization, the implementation further applies a second-moment normalization on the matrix update magnitude before weight decay and the learning-rate step. In that sense, our project studies coefficient schedules inside the specific normalized Muon variant actually used by this training stack. This detail is important for interpreting the results, because the measured update geometry is the geometry of the real implemented search direction rather than of an idealized Muon formula.

For the spectral geometric study, every 10M training tokens we sample representative update matrices and record singular-value statistics together with a short-side semi-orthogonality error:

$$\varepsilon(X) = \begin{cases} \|X^\top X - I\|_F, & X \text{ tall or square,} \\ \|XX^\top - I\|_F, & X \text{ wide.} \end{cases} \quad (7)$$

This metric is applied to `buffer_post`, `g_pre`, and `g_post`. It allows us to distinguish between geometry inherited from the training trajectory before orthogonalization and geometry created directly by the NS map after orthogonalization.

3 Main Results

3.1 Benchmark Results

At fixed depth, the four Muon schedules have nearly the same wall-clock cost. Table 1 shows that all five-step Muon schedules finish in about 1132 seconds. This is an important result: once the number of NS steps and the surrounding optimizer path are fixed, coefficient design barely affect training throughput.

Table 1: Benchmark wall-clock and peak memory for 100M-token runs.

Method	Wall-clock (s)	Peak memory (MiB)
AdamW	1063.99	8380
Stable ($T = 5$)	1133.54	8074
Manual ($T = 5$; $F3_S2$)	1132.34	8074
Fast ($T = 5$)	1132.71	8074
PE ($T = 5$)	1132.58	8074

The main cost difference is the orthogonalization path, not the choice of five-step coefficients. In our implementation, both optimizers are hand-written in PyTorch, so the comparison is not confounded by different kernel maturity. AdamW is faster because it avoids the matrix multiplications and normalization operations required by NS orthogonalization. Within the $T = 5$ Muon family, the coefficients change the scalar polynomial but not the number of polynomial evaluations, so the wall-clock cost stays close. As a result, the practical tradeoff is between additional geometric correction and the amount of validation-loss improvement obtained from it.

3.2 Training Results

Muon consistently improves pretraining quality over the AdamW baseline under the same token budget. Table 2 shows that every Muon variant reaches lower final validation loss and lower validation-loss AUC than AdamW within 100M training tokens. This is the most basic empirical result of the project: introducing semi-orthogonalized matrix updates changes optimization behavior enough to produce a substantial quality gain in this controlled transformer-pretraining setting.

Table 2: Core $T = 5$ training metrics. Mean final loss is reported when multi-seed follow-up is available; otherwise the single-run final loss is shown. AUC is token-normalized, so lower values indicate a better average trajectory.

Method	Mean final loss	Std.	AUC
AdamW	5.3737	–	5.9677
Stable ($T = 5$)	4.5273	–	5.3166
Manual ($T = 5$; $F3_S2$)	4.3176	–	5.0295
Fast ($T = 5$)	4.2892	0.0056	4.9340
PE ($T = 5$)	4.2902	–	4.9338

Fast and Polar Express give the strongest training results in the fixed-depth $T = 5$ comparison. As shown in Figure 1 and Table 2, Fast and PE are numerically the two strongest schedules at fixed depth: Fast achieves the best mean final loss, while PE is extremely close and gives the best AUC. Manual improves clearly over Stable, but it does not match the top two aggressive five-step schedules. The multi-seed follow-up also suggests that this result is reasonably robust: for Fast ($T = 5$), the standard deviation of final loss is about three orders of magnitude smaller than the mean.

Ablation studies on Iteration depth and the PE lower bound To keep this distinction explicit, Tables 3a and 3b summarize these follow-up sweeps separately from the fixed-depth comparison in Table 2.

Within the Manual family, moving from the $T = 5$ hybrid point to the deeper $T = 7$ – 10 range yields a clear one-step improvement in final loss, but the deeper settings then cluster tightly around 4.29. This pattern suggests diminishing returns from additional NS depth once the schedule is already strong enough to reshape the small-singular-value region aggressively. Deeper schedules are modestly slower, which is expected because they execute more NS iterations.

Within the PE family, the lower-bound parameter produces a non-monotonic tradeoff. The strongest final loss appears at $\ell = 3 \times 10^{-3}$, while the best AUC appears at $\ell = 10^{-3}$, and pushing the lower bound down to 3×10^{-5} degrades both metrics. This is consistent with the spectral-map interpretation: changing ℓ changes the interval on which the PE polynomial is designed, but stronger correction at very small singular values is not automatically rewarded by the language-model objective.

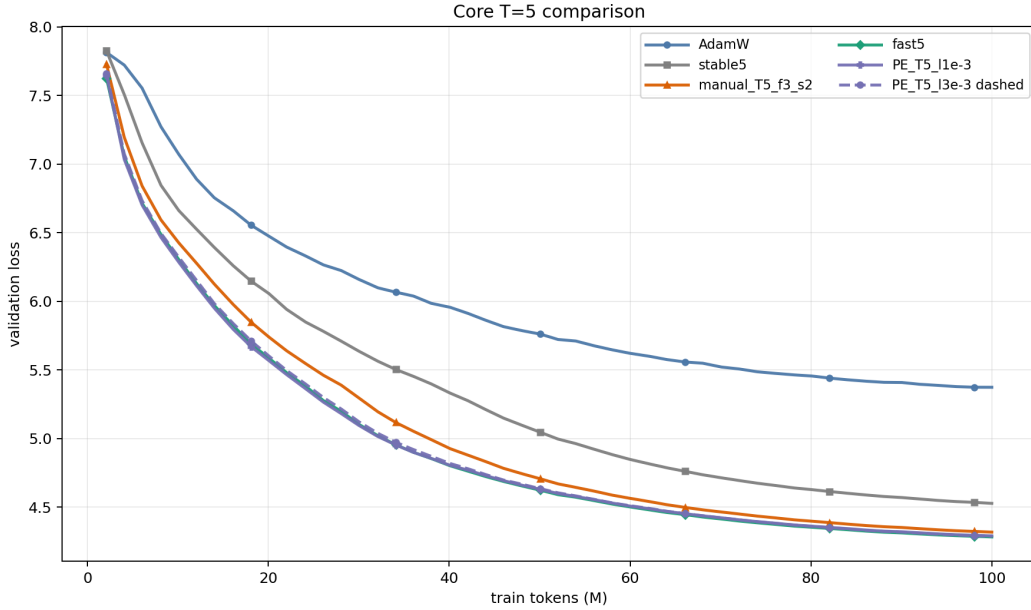


Figure 1: Core $T = 5$ validation-loss curves. The comparison isolates coefficient-sequence effects at fixed depth.

(a) Manual-depth ablation.				(b) PE lower-bound ablation at $T = 5$.			
Schedule	Final loss	AUC	Time (s)	ℓ	Final loss	AUC	Time (s)
$T = 5$ ($F3, S2$)	4.3176	5.0295	1095.57	3×10^{-3}	4.2873	4.9462	1095.88
$T = 7$ ($F4, S3$)	4.2873	4.9312	1107.66	1×10^{-3}	4.2902	4.9338	1094.63
$T = 8$ ($F5, S3$)	4.2890	4.9253	1119.07	3×10^{-4}	4.2998	4.9349	1098.28
$T = 9$ ($F3, S6$)	4.2887	4.9271	1118.57	3×10^{-5}	4.3125	4.9479	1094.97
$T = 9$ ($F4, S5$)	4.2923	4.9268	1118.14				
$T = 9$ ($F5, S4$)	4.2947	4.9286	1118.87				
$T = 9$ ($F9, S0$)	4.2971	4.9308	1120.00				
$T = 10$ ($F5, S5$)	4.2949	4.9282	1131.78				

Table 3: Single-seed ablation studies under the same 100M-token budget. These sweeps probe specific Muon hyperparameters.

3.3 Spectral Geometry Results

Newton–Schulz schedules create visibly different update geometries. Table 4a aggregates the sampled semi-orthogonality errors. The error drops sharply from **g-pre** to **g-post**, confirming that the NS map is doing the intended geometric work. At the same time, the schedules remain clearly separated after orthogonalization: deeper Manual and PE schedules produce smaller **g-post** error than Fast, even though their final validation losses are higher.

Schedule sensitivity is larger in attention layers than in MLP layers. The right side of Figure 2 and Table 4b show that attention matrices are much more sensitive to schedule choice than MLP matrices. For example, PE at $T = 9$ has attention **g-post** error 0.0481, while its MLP error is still about 0.7081. This suggests that the schedule-dependent spectral effect is not uniform across the Transformer; attention projections are the clearest location where improved singular-value control appears.

The scalar maps explain where the geometry difference comes from. For a schedule with T NS steps, define the composed scalar map p_T by repeatedly applying $p_k(\sigma) = a_k\sigma + b_k\sigma^3 + c_k\sigma^5$. Figure 3 first shows the one-step behavior: the Fast coefficient map amplifies small singular values much more strongly than the stable NS map, but it also has a region with negative derivative. This is why Fast behaves less like a conservative contraction and more like an aggressive spectral reshaping rule.

(a) Mean semi-orthogonality errors.

Method	buffer_post	g_pre	g_post
Vanilla	5.0730	5.1962	0.9006
Fast	3.4080	3.4584	0.6444
Manual ($T = 5$)	3.2840	3.3425	0.6866
Manual ($T = 9$)	1.9864	2.0471	0.4431
PE ($T = 9$)	2.1216	2.1781	0.3781

(b) Mean $\mathbf{g_post}$ error by module.

Method	Attention	MLP
Fast	0.5146	0.7743
Manual ($T = 9$)	0.1776	0.7086
PE ($T = 9$)	0.0481	0.7081

Table 4: Spectral summaries for overall and module-specific geometry. Lower semi-orthogonality error means the update is closer to the polar-factor target.

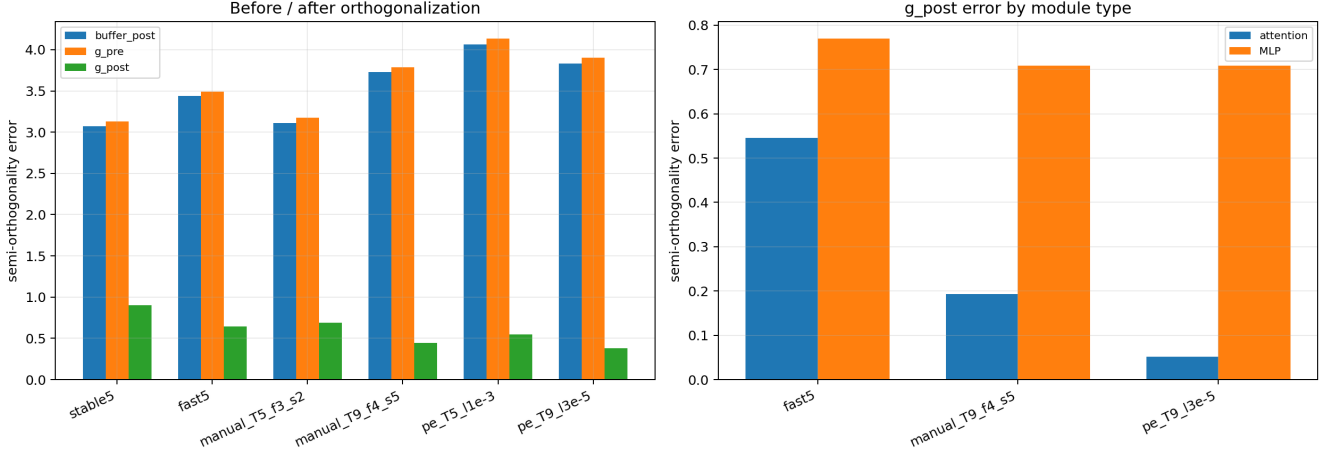


Figure 2: Spectral diagnostics from sampled update matrices. Left: error before and after the NS map. Right: module-level $\mathbf{g_post}$ error for representative schedules.

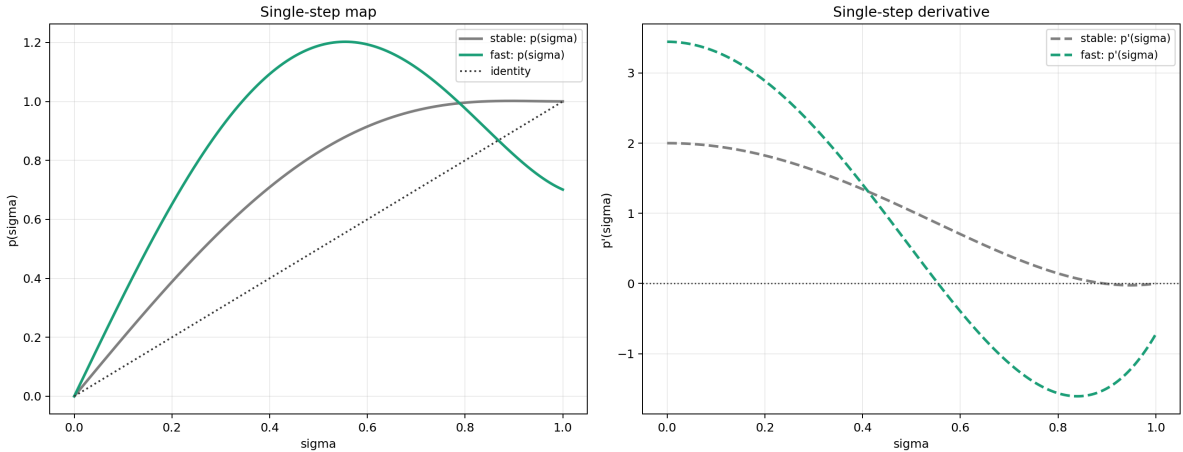


Figure 3: Single-step Newton-Schulz maps and derivatives for the stable and fast coefficient choices. The derivative panel shows where the map is locally amplifying or reversing order in singular-value space.

After several steps, these local differences accumulate into very different composed maps. Figure 4 shows that Vanilla moves small singular values slowly, Fast amplifies them more aggressively, and deeper Manual or PE schedules can push them much closer to the polar target. Figure 5 adds the derivative of the composed map. Large or oscillatory derivatives indicate that the final mapped value is highly sensitive to the input singular value, which helps explain why stronger orthogonalization is not automatically a better training update.

Table 5: Small-singular-value amplification implied by the composed maps.

Method	$p_T(10^{-5})$	$p_T(10^{-5})/10^{-5}$
Vanilla	3.20×10^{-4}	32
Fast	4.85×10^{-3}	485
Manual ($T = 9$)	4.50×10^{-2}	4502
PE ($T = 9$)	7.13×10^{-1}	71309

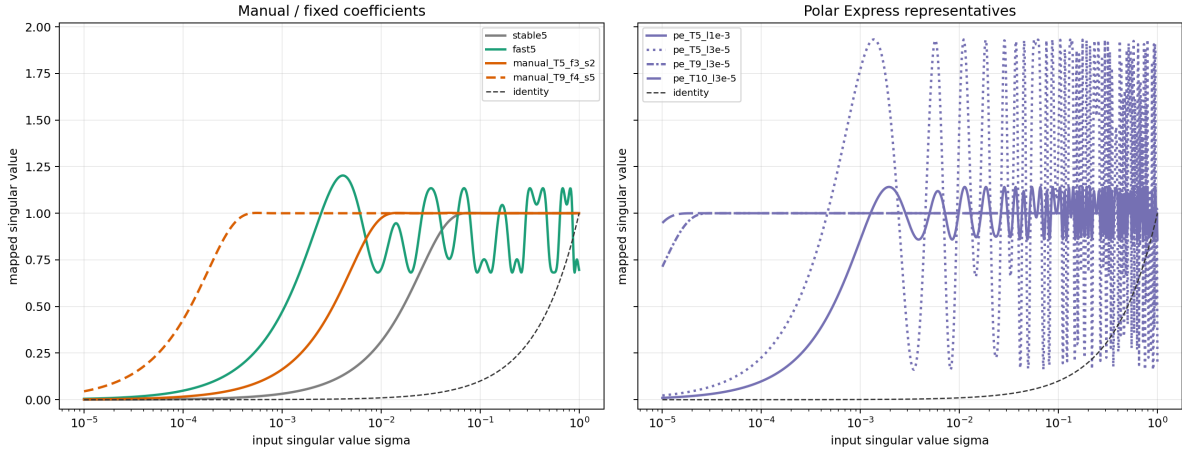


Figure 4: Composed singular-value maps. The dashed identity line is the input singular value σ ; the colored curves show the value after applying a full schedule.

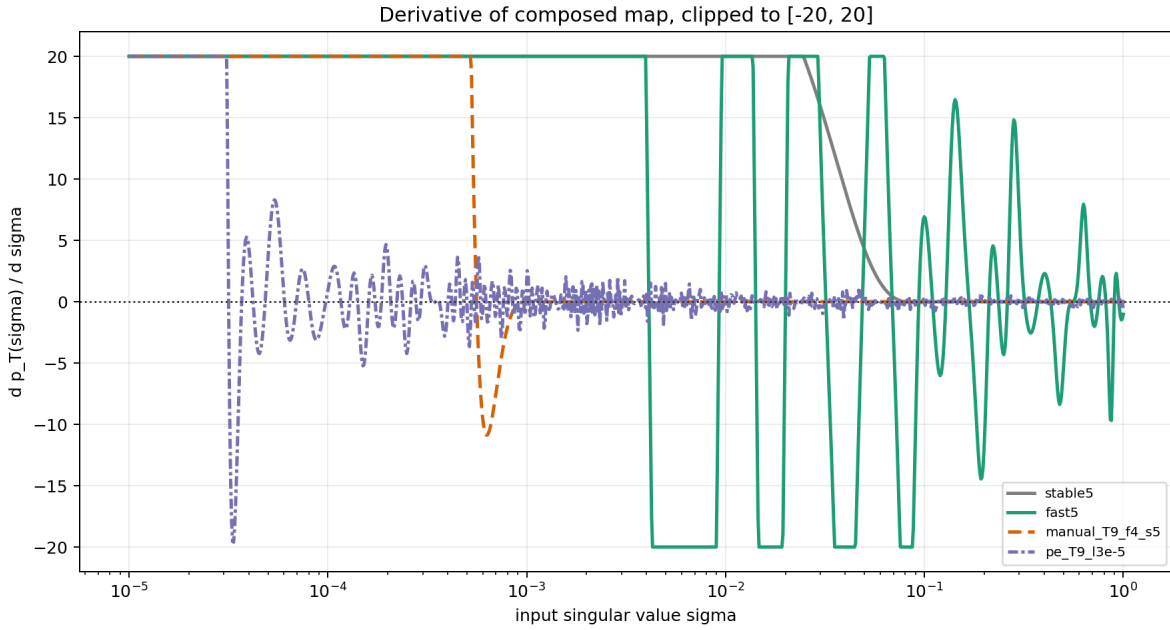


Figure 5: Derivatives of the composed singular-value maps. The curves describe how sensitively the final mapped singular value changes with respect to the input singular value.

The amplification ratio in Table 5 should be interpreted together with the mapped value. For example, PE at $T = 9$ has a gain of about 7.1×10^4 at 10^{-5} because it maps that singular value to about 0.713, not because it pushes it beyond the polar target by the same factor. The lower-bound parameter in PE is therefore not a cosmetic hyperparameter: it changes the singular-value interval for which the polynomial is designed. This gives a natural explanation for the training results. Stronger spectral correction produces cleaner post-orthogonalization geometry, but the resulting update can also

be more sensitive to the current spectrum of the stochastic gradient.

4 Conclusion

This project asked how different Newton–Schulz coefficient schedules change the spectral geometry of Muon updates, and how those geometric changes relate to transformer pretraining behavior under a fixed experimental stack. The main conclusion is that the schedule matters because it changes the geometry of the actual matrix update, not merely because it changes an implementation detail inside the optimizer. At fixed depth, fast Muon and well-chosen PE schedules are much stronger than the stable NS schedule. This indicates that aggressive small-singular-value amplification is useful in the early-scale pretraining regime studied here.

The spectral evidence gives a more detailed picture. Different schedules produce different composed singular-value maps, and those maps are reflected in the measured geometry of real update matrices. Deeper Manual and PE schedules substantially reduce post-orthogonalization error, especially in attention layers. However, this cleaner geometry does not translate monotonically into lower validation loss: in the representative multi-seed runs, Fast remains the strongest practical baseline at 100M tokens, while the more orthogonal $T = 9$ schedules are close but not better.

Overall, the experiments suggest a balanced view of NS schedule design in Muon. Fast coefficients are highly effective for the present small-transformer pretraining setting, while Manual and PE schedules provide a more controllable way to reshape the spectrum and study the geometry of the update. The most interesting mathematical question left by the experiments is not simply how to make the update more orthogonal, but how the composed singular-value map should trade off amplification, sensitivity, and stability for the spectra that actually occur during training.

References

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